

DBSP

THE HIGH PERFORMANCE PUZZLE

Strength & Conditioning for Ultimate Frisbee

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CHAPTER 01



THE PERFORMANCE PUZZLE

Why S&C Belongs in Your Frisbee Program

Before we get into what to do, it helps to understand what your body is actually being asked to handle.

GPS data collected at the international level gives us the clearest picture we have of what frisbee demands from athletes who are genuinely prepared for it. In a typical game at international level, a player covers around 3,500 metres, completes close to 50 acceleration efforts, and logs over 750 metres at high speed. At the elite ceiling — the kind of game you would face deep in a World Ultimate Championships bracket — those numbers push to 5,000 metres of total distance, over 1,000 metres at high speed, 343 metres at very high speed, and 74 acceleration efforts across a single game.

WUC - MEDIAN GAME DEMANDS

3,479m Game distance	759m High speed run	226m Very high speed	50 Accelerations
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But the per-point breakdown is where it gets interesting.

At WUC level, every single point costs you roughly 265 metres of total running, 54 metres at high speed, and 18 metres at very high speed — alongside nearly 4 acceleration efforts and 2.4 deceleration efforts. Those numbers don't change much whether you're playing at club or international level. What changes is how many times you repeat them. A peak game pushes to 38 minutes of live point time and beyond. The difference between a well-prepared athlete and an underprepared one isn't the effort they put into each point — it's whether they can maintain that output from the first point to the last, game after game across a tournament day.

WUC - PER POINT DEMANDS

265m Total dist / point	54m HSR / point	18m VHSR / point	3.9 Accels / point
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WUC - PEAK GAME CEILING

5,026m Peak game distance	1,035m Peak HSR	343m Peak VHSR	74 Peak accelerations
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That's the problem strength and conditioning is built to solve.

S&C's primary purpose is to develop the physical qualities that transfer directly to on-field performance — your ability to sprint, cut, contest aerially, and execute under fatigue. The secondary purpose is reducing injury risk and extending playing longevity. An athlete who has trained the demands of their game is simply harder to hurt and slower to break down.

Performance in frisbee isn't one thing. It's the product of six interconnected pillars: strength and conditioning, on-field training, technical and tactical development, psychology and mental skills, nutrition, and recovery. S&C; doesn't replace any of the others — it creates the physical platform that allows all of them to actually work. A player who can't maintain speed and power output in the second half of a long game isn't just physically limited — their decision-making suffers, their technical execution breaks down, and their confidence takes a hit.

The question this guide is built to answer isn't whether physical preparation matters. The data makes that clear. The question is how to build that preparation in a way that's specific to frisbee, realistic to manage alongside training and life, and grounded in what actually works.

CHAPTER 02

BUILT FOR FRISBEE

The Five Physical Qualities and How They Show Up on the Field



Most gym programs are built around generic outcomes — lose weight, get stronger, look better. That's fine for general health, but it's not what frisbee athletes need. If your physical preparation isn't connected to what the game actually demands, you're spending time and energy on the wrong things.

Frisbee performance is built on five physical qualities. Each one has a direct application to something that happens on the field.

SPEED — is the quality that determines how much of the field you can threaten. Big space cutting — the deep ball, the clearing run, the break-side continuation — is fundamentally a speed problem. If your opponent can sit on your first step and not worry about getting beat long, your cutting options narrow immediately. Speed also determines your ability to recover defensively and contest plays downfield that you have no business getting to unless you're fast.

POWER — is speed applied to a specific moment. Aerial contests, layouts, explosive first steps off the mark — these are power expressions. Power is what separates the athlete who gets to the disc from the one who also comes down with it. It's trainable, and it responds well to the right gym work.

AGILITY — is the quality that lives in the small spaces. Handler footwork, tight cuts in a congested endzone, staying with your player when they change direction — agility is where frisbee is actually played most of the time. Speed matters, but agility is what makes speed usable in a real scenario where there's almost never a clear straight line.

CAPACITY — is the engine underneath everything else. It's the physiological foundation that determines whether your speed, power, and agility are still available in the second half of a long game, or whether they've quietly dropped off while you're telling yourself you're fine. The per-point demands of frisbee are consistent, but the game keeps asking for them over and over. Capacity is what allows you to keep delivering.

MOBILITY — is the quality that doesn't get enough attention until something breaks down. Throwing mechanics rely on shoulder, thoracic, and hip mobility. Top speed mechanics require hip flexor length and ankle range. Change of direction is significantly impacted by mobility restrictions that most players have but haven't addressed. Mobility won't make you faster overnight, but a lack of it will quietly put a ceiling on every other quality you develop.

These five qualities don't exist in isolation. A well-designed S&C; program develops all of them in proportion to the demands of the sport — not equally, not arbitrarily, but in the ratios that frisbee actually requires. That's the difference between training hard and training smart.

CHAPTER 03



THE ADAPTATION CURVE

How Training Actually Works (and Where Most Players Get It Wrong)

Every training session you do produces one of three outcomes. Your fitness improves, it stays the same, or it declines. Which one you get isn't random — it's determined by whether the training stimulus you applied was appropriate for where your body currently is.

Understanding this is the difference between a program that works and one that just keeps you busy.



The underlying mechanism is straightforward. When you apply a training load to the body — a gym session, a hard field session, a conditioning run — you create fatigue and a small amount of damage to the systems being trained. Given adequate recovery, the body doesn't just repair to where it was. It overcompensates slightly, rebuilding to a marginally higher level of capacity than before. Do that repeatedly, with progressive increases in the stimulus, and fitness improves. That process is called supercompensation, and it's the biological basis of every training adaptation that has ever occurred.



Supercompensation only works within a specific window. Apply too little stimulus and the body has no reason to adapt — fitness plateaus or slowly declines. Apply too much, too fast, without adequate recovery, and the body can't complete the adaptation cycle. The

central question of training program design is finding the load that sits between those two failure modes: enough to drive adaptation, not so much that it outpaces recovery.

For frisbee athletes this plays out in a particular way. Most players either train inconsistently — doing too little for too long, then spiking their load in the weeks before a big tournament — or they add gym work on top of an already full schedule without accounting for the cumulative effect. Both patterns produce predictable outcomes: stalled fitness in the first case, injury or burnout in the second.

The adaptation curve doesn't care about your intentions. It responds to what you actually do, and how well you recover from it. A player who trains at a moderate, consistent load with good recovery will outperform a player who trains harder but inconsistently — because adaptation requires the full cycle to complete.

CHAPTER 04



WHAT DRIVES ADAPTATION

The Four Levers That Determine Whether Your Training Actually Sticks

Training is only one part of getting fitter. Most athletes treat the session as the entire equation — if they worked hard, they expect results. What actually determines whether adaptation occurs is the relationship between four things: training, on-field load, nutrition, and recovery. Pull any one of them without accounting for the others and the system breaks down.

THE FOUR LEVERS

Training provides the stimulus	On-field cumulative stressor	Nutrition raw material for repair	Recovery where adaptation happens
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Each lever contributes to whether the body has the right conditions to adapt. Training provides the stimulus. On-field sessions add to the cumulative load — sometimes more than athletes realise. Nutrition provides the raw material the body needs to repair, rebuild, and fuel the next session. Recovery is the time and process through which the adaptation actually happens.

The lever most commonly neglected is nutrition. Athletes will structure their training carefully and prioritise sleep, but chronically underfuel — either generally, or specifically around sessions — and then wonder why their training isn't progressing. Training is a demand, and nutrition is the supply. Mismatch the two and adaptation suffers.

The second most neglected lever is recovery. Not because athletes don't value rest, but because recovery is often passive — it happens between sessions, so it doesn't feel like doing something. The decisions you make in the hours after a hard session directly determine how much of the adaptation stimulus you're able to absorb.

Everything in the performance puzzle feeds into one outcome: the ability to play and train better. The athletes who make the most consistent progress aren't always the ones training hardest. They're the ones managing all four levers well, most of the time.

CHAPTER 05



ON-FIELD TRAINING

Why Your Frisbee Training and Your Gym Work Have to Talk to Each Other

On-field training is not a free hit. This is one of the most common misunderstandings in how frisbee athletes manage their preparation — the idea that gym work is the training load and frisbee sessions are just frisbee. In reality, every field session is a physiological stressor that sits on top of whatever else you've done that week, and your body doesn't distinguish between them when it's deciding how much recovery it needs.

The purpose of on-field training is to develop the ability to handle the specific demands of the game — the tactical patterns, the technical skills, the movement vocabulary of frisbee. Done well, it builds those things while also contributing positively to physical preparation. Done poorly, or without awareness of where it sits in the broader training week, it competes with recovery and undermines the adaptation you're trying to drive in the gym.

This relationship runs in both directions. S&C; supports on-field training by building the physical foundation that allows athletes to train at higher intensity, absorb more volume, and recover faster. But on-field training can hamper S&C; if the scheduling creates cumulative fatigue that prevents quality gym work.

Frisbee always takes priority. When there's a conflict between gym work and field training, the field wins — because technical and tactical development requires presence and quality that a fatigued athlete can't reliably deliver. But that priority doesn't mean the gym disappears. It means the program is structured so that conflicts are rare, and when they do arise, the response is intelligent rather than reactive.

The weekly templates in Chapter 9 show how this balance looks in practice.

CHAPTER 06



RECOVERY

Cold, Heat, Compression — What Actually Works and When to Use It

Recovery is not one thing. It's a set of tools, each with a specific context where it's most effective — and using the wrong tool at the wrong time either does nothing or actively works against you. The most common example is the athlete who cold plunges after every single session regardless of what that session was. Cold exposure has real value, but it's not a universal recovery solution, and applying it indiscriminately is a sign of following trends rather than understanding the physiology.

RIGHT TOOL, RIGHT CONTEXT

**Heat +
movement**
after training

**Cold +
compression**
after games / tournaments

Sleep
the non-negotiable

AFTER TRAINING SESSIONS — the priority is enhancing blood flow, clearing metabolic byproducts, and beginning the repair process. Heat and light movement are the right tools here. The goal is to keep circulation active without suppressing the adaptation stimulus. Some post-training inflammation is part of the process — aggressive cold exposure immediately after a gym session can blunt the very adaptation you were trying to drive.

AFTER GAMES AND TOURNAMENTS — the priority shifts from adaptation to restoration — getting the body ready to perform again as quickly as possible, often within hours. Cold exposure reduces inflammation, limits soft tissue swelling, and accelerates the recovery of performance capacity. Compression supports circulation and lymphatic clearance. At tournaments specifically, cold and compression aren't optional extras — they're part of the performance plan for day two.

Sleep is the element that underpins all recovery modalities. No ice bath, compression garment, or recovery session replaces what happens physiologically during quality sleep. Growth hormone release, tissue repair, neural recovery, and psychological restoration all occur primarily during sleep. An athlete who uses every recovery tool available but consistently sleeps six hours is leaving more on the table than any modality can give back.

Recovery can be passive — and often should be. Doing less is recovering. The temptation to fill every post-session window with another modality works against the process. Sometimes the most effective recovery strategy is simply to stop, rest, and let

the body do what it's designed to do.

CHAPTER 07



NUTRITION

Fuel, Build, Recover — How to Eat for the Demands of Frisbee

Nutrition does three things for a frisbee athlete. It fuels training and competition. It supports the physiological processes that drive adaptation — hormone production, tissue repair, immune function. And it accelerates recovery between sessions and games. Understanding them separately helps clarify why what you eat, and when you eat it, matters as much as how much you eat.

FUELLING — is the most intuitive role. Carbohydrates are the primary fuel source for high-intensity intermittent sport — exactly the profile of frisbee. An athlete who arrives at training or competition with depleted carbohydrate stores will feel it in their speed, their decision-making, and their ability to maintain output. Chronically underfuelling carbohydrate is one of the most common and most impactful nutritional mistakes frisbee athletes make.

BUILDING — is the role protein plays most directly. Protein provides the amino acids required for muscle repair and rebuilding after training stress. It also supports immune function, which takes a hit during heavy training blocks and tournaments. Getting adequate protein in the hours after a hard session accelerates the repair process and positions the body better for the next training stimulus.

RECOVERY NUTRITION — is where the balance between carbohydrate and protein becomes context-dependent. After a field session where carbohydrate stores have been depleted, the priority is replenishment. After a strength session, protein moves up the priority order. In practice, most recovery meals should contain both — the ratio is what shifts.

Hydration sits underneath all three roles. Even mild dehydration measurably impairs speed, power output, decision-making, and thermoregulation. Frisbee is predominantly an outdoor sport played in warm conditions, often across long tournament days. Hydration isn't a game-day concern — it's a daily habit that determines how well everything else in your nutrition plan actually works.

CHAPTER 08



THE MENTAL EDGE

Why Psychology Isn't Soft — and Why Fitness is the Foundation of It

High performance is a mindset. That's not a motivational poster sentiment — it's a description of something that requires active, ongoing attention and development. The psychological demands of competitive frisbee are real: pressure situations, tight scorelines, errors in front of teammates, the cumulative mental weight of a long tournament day. How an athlete responds to those demands is a skill, and like every other skill, it can be trained.

But here's what often gets missed: **the foundation of psychological performance is physical.** The mental edge doesn't exist independently of the body it runs on.

Sleep and nutrition are the two most direct physical inputs into psychological function. Sleep deprivation impairs reaction time, emotional regulation, decision-making, and the ability to maintain focus under pressure — all before an athlete has thrown a single disc. Chronic underfuelling does the same thing more slowly, with athletes often attributing the mental flatness and irritability to stress or motivation rather than its actual cause.



"Pressure is not the enemy. It's the price of entry to high performance."

Physical preparedness directly affects confidence. An athlete who has genuinely prepared for the demands of the game arrives at competition in a different mental state than one who hasn't. That confidence isn't manufactured. It's earned through preparation, and it's one of the most durable psychological assets an athlete can have going into a high-pressure situation.

The athletes who perform best under pressure aren't the ones who feel it least — they're the ones who have built the physical and psychological capacity to function well while feeling it. That capacity is built in training, through deliberate exposure to uncomfortable situations and the accumulated evidence of preparation done properly.

The mental edge is real, trainable, and worth developing deliberately. But it sits on top of a physical foundation, and that foundation has to be built first.

CHAPTER 09



BUILDING YOUR WEEK

A Practical Template for Training Weeks and Match Weeks

Understanding the principles behind training is one thing. Fitting them into a real week — alongside field sessions, work, and everything else — is where most athletes struggle. The templates below give a structural framework that puts the right sessions in the right order and protects the qualities that matter most.

NON-NEGOTIABLE RULES

6 hrs min

between any two sessions

Field first

frisbee always takes priority

3 gym/wk

standard training week

Training Week

Anchored by three gym sessions — Monday, Wednesday, and Friday — with field training on Tuesday and Thursday. Monday's afternoon conditioning session sits at least six hours after the morning gym session. Saturday's conditioning and skills session rounds out the week before a full rest day on Sunday.

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
AM	Gym	—	Gym	—	Gym	Cond & Skills	—
PM	Cond & Skills	Training	—	Training	—	—	Rest

Match Week

Three gym sessions early in the week — Monday, Tuesday, and Thursday — with the game on Saturday. Field sessions Tuesday and Thursday are retained but managed for intensity. There is no gym work in the 48 hours before competition.

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
AM	Gym	Gym	—	Gym	—	Game	—
PM	Cond & Skills	Training	—	Training	Travel / Prep	—	Rest

Morning gym sessions work best when the athlete has eaten adequately beforehand. Field sessions in the evening should be followed by a recovery meal before sleep. Rest

days are non-negotiable — they're when the adaptation from the week's work actually consolidates.

CHAPTER 10



RECOVERY IN PRACTICE

When to Use What — Mapped to Your Week

Knowing which recovery tools work is one thing. Knowing when to deploy them across a real training week is what makes the difference between recovery as a concept and recovery as a practice. The schedule below maps modalities to the weekly templates from Chapter 9.

MONDAY — Follows a rest day, so the body arrives relatively fresh. The afternoon conditioning session after the morning gym session means Monday carries a double training load — nutrition first, sleep second in the evening. No additional modalities needed unless there's residual soreness.

TUESDAY — Field training day. Post-session priority is getting a recovery meal in promptly and managing any soft tissue soreness. Light heat exposure supports blood flow without suppressing the training stimulus.

WEDNESDAY — Carries a morning gym session and is otherwise relatively light. Good day for deliberate recovery work — mobility, soft tissue maintenance, heat if available. The body needs to be prepared for Thursday's field session.

THURSDAY — Mirrors Tuesday. Post-session nutrition is the priority, followed by sleep. If the week has been heavy, this is the evening to take recovery seriously.

FRIDAY — In a training week, the third gym session — keep the evening clear and prioritise sleep. In a match week, Friday is travel and preparation — hydration becomes a specific focus, particularly if flying. The flight hydration protocol from Chapter 11 applies directly.

SATURDAY — In a training week: conditioning and skills in the morning. In a match week: game day. Post-game recovery shifts to cold and compression — get cold exposure in within an hour of the final point, use compression on the legs overnight, and prioritise a full recovery meal before sleep.

SUNDAY — Full rest day. Passive recovery — do less, move less, eat well, sleep long. In a tournament context with a second game day, replicate the Saturday post-game protocol exactly and add an emphasis on carbohydrate restoration overnight.

The through-line across all of it is consistency. A single ice bath the night before a tournament doesn't undo a week of poor sleep and inadequate nutrition. The athletes

who recover best aren't the ones with the most sophisticated protocols — they're the ones who nail the basics, every day, without making it complicated.

CHAPTER 11

GAME DAY NUTRITION

The Full Timeline — From the Night Before to Half Time

Game day nutrition isn't a single decision — it's a sequence of decisions that starts the night before and runs right through to the last point. Get the sequence right and you arrive at the first point fuelled, sharp, and hydrated. Get it wrong and you're chasing your tail from the opening point, trying to recover energy and focus that should have been banked hours earlier.

The timeline below is built around a 3:30pm game — typical for a single-game-per-day format such as WUC. Adjust the timing proportionally for earlier or later starts.

NIGHT BEFORE — Carbohydrate-dominant dinner — pasta, rice, potatoes — followed by a post-dinner snack or smoothie. Hydration is an active focus from this point. The goal going to bed is full glycogen stores and good fluid status.

8:30-9:30AM — Large breakfast. Eggs, bacon, toast, fruit, smoothie. This is the last full meal before the game and it needs to be substantial. Don't skip breakfast on game day because you're not hungry — appetite suppression before competition is normal.

11:30AM-1:30PM — Pre-game meal. Two to three hours before game time. A ham or chicken toastie, a hot meal, pancakes, fruit — something that sits comfortably and continues to load carbohydrate. Not the time for anything new or unfamiliar.

2:30PM — Pre-warm-up. Caffeine if you use it — pre-workout or caffeine gummies — plus a light snack or gel. Goal: alert, fuelled, nothing heavy in the stomach.

3:15PM — Post warm-up, pre-pull. A gel, lollies, and Powerade or electrolytes. Top up caffeine if needed. Non-negotiable regardless of how you feel. By the time hunger arrives during a game, you're already playing catch-up on fuel.

HALF TIME — Gel, lollies, electrolyte top-up. Keep it simple and fast. If it's hot, prioritise fluids. Maintain blood glucose and fluid levels through the second half.

Multi-game days — club tournaments and pool play

At tournaments where you're playing two or three games in a day, the 3:30pm timeline doesn't apply directly. For morning games, the large breakfast becomes the pre-game meal. The pre-warm-up snack and gel protocols still apply — just tighten the timing to fit the earlier start. Between games, the priority is rapid carbohydrate restoration — eat as

soon as possible after the final point and don't wait until you're hungry.

The core rules — load carbohydrate early, time caffeine deliberately, use gels and electrolytes around warm-up and at the break — apply regardless of how many games you're playing. The timeline adjusts, the logic doesn't.

Travel and flight days

Air travel is dehydrating — cabin air is extremely dry and fluid losses are higher than they feel. The protocol: at least 600ml of water per hour awake on travel days, electrolytes between meals, chewing gum or mints to maintain airway moisture, and a nasal spray if flying long-haul. Arriving at a tournament already dehydrated is one of the most avoidable performance mistakes in team sport.

CHAPTER 12



SUPPLEMENTS

What Actually Works, What It Does, and How to Use It

The supplement industry is enormous, poorly regulated, and built on the premise that more is better. Most of what's marketed to athletes either doesn't work, works only marginally, or works in contexts that don't apply to team sport. This chapter covers the supplements with genuine evidence behind them for frisbee athletes — what each one does, when to use it, and why it earns its place.

Everything else can stay on the shelf.

WHEY + CARBOHYDRATE — supports muscle repair and recovery after both gym and field sessions. For post-game recovery specifically, a whey and carbohydrate mix hits both restoration priorities simultaneously. On tournament days where the window between games is short, a whey and carbohydrate shake taken straight after the final point is the closest practical substitute for a full meal. Look for roughly two to three parts carbohydrate to one part protein.

CREATINE — is the most evidence-backed performance supplement available. It enhances the body's capacity for repeat high-intensity efforts — exactly the physical profile of frisbee. Sprint efforts, acceleration bursts, repeated cutting sequences — creatine supports the energy system that powers all of them. Take 3-5 grams daily, timing is not critical. Benefits accumulate over weeks of consistent use rather than arriving immediately.

CAFFEINE — is a well-established performance enhancer for speed, power, reaction time, and endurance. Effective dose: approximately 4mg per kilogram of bodyweight, taken 30-45 minutes before the performance window. Pre-workout formulas and caffeine gummies are the most controllable delivery formats for game day. Tolerance builds with daily use — time it deliberately to preserve its effect when it matters most.

ENERGY GELS — deliver concentrated carbohydrate in a portable, fast-absorbing format. One gel provides approximately 30 grams of carbohydrate — roughly equivalent to 400ml of Powerade. They're a top-up tool for maintaining blood glucose during competition, used at the pre-pull window and at half time as outlined in Chapter 11.

ELECTROLYTES — restore the sodium, potassium, and magnesium lost through sweat. Plain water rehydrates but doesn't replace electrolytes, and relying on water alone during heavy sweating can dilute blood electrolyte concentration. Electrolyte products should be the hydration tool of choice during tournament days — not just for the first

game but across the full day.

The data that underpins the targets and thresholds in this guide comes directly from GPS monitoring of elite frisbee competition — the same data that powers Frisbee Track, DBSP's training load monitoring app built specifically for Ultimate Frisbee. The app translates those match demands into personalised weekly targets and prescriptions for any player, at any level, without requiring access to GPS hardware. If this guide has given you the framework, Frisbee Track gives you the numbers to train against.

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